1 mit media lab 2 BROWN 3 UNIVERSITY OF TORONTO

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HapticHearing: A Haptic Feedback System for Complementing Speech Perception

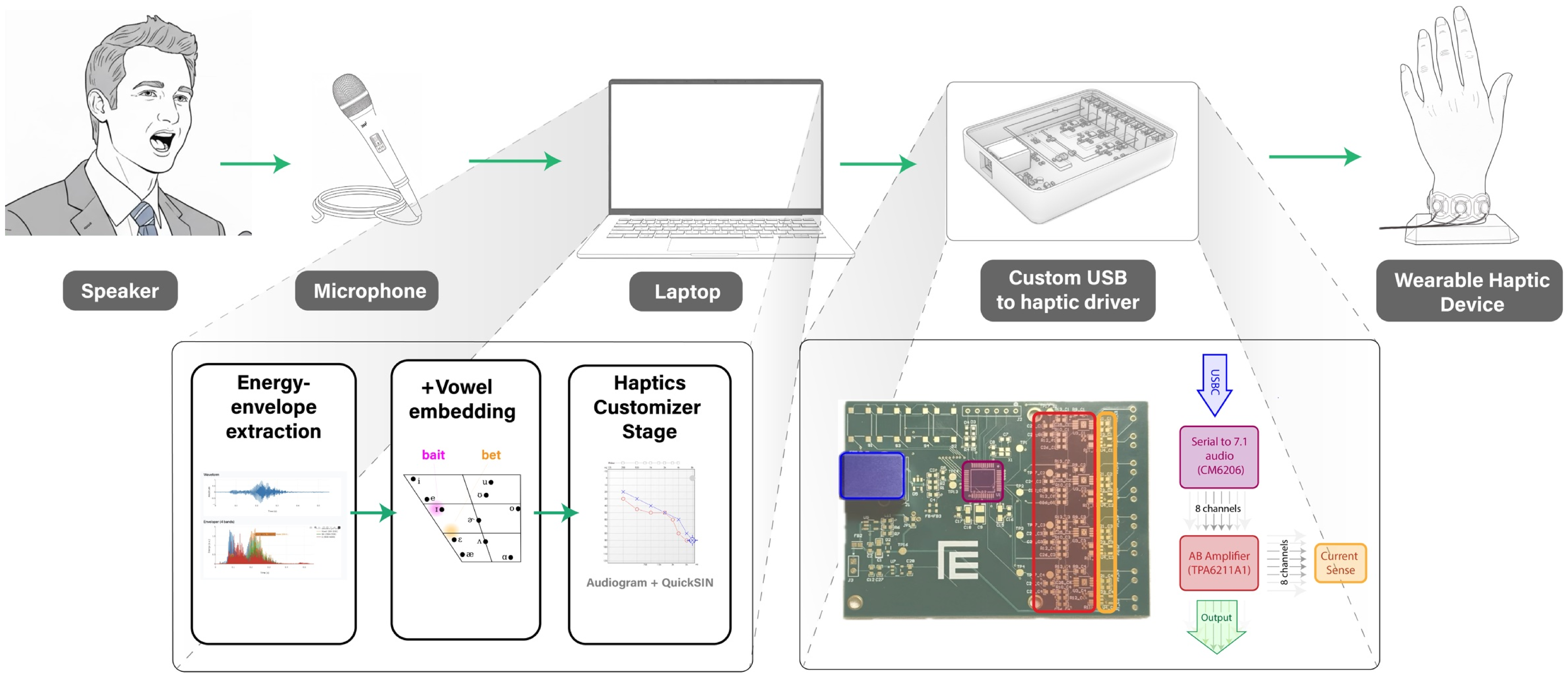
Contributions

· Developed a custom, **multi-actuator haptic device** in three wearable form factors (bracelet, necklace & over-ear).

· Conducted a **psychophysical study** (n=9) to validate vibration detection thresholds and spatial localization accuracy across the different designs.

· Developed an end-to-end software pipeline that performs real-time audio analysis and haptic rendering, translating phonemic information into spatiotemporal patterns.

HapticHearing Overview



Future Work

· We are actively **seeking feedback** on which auditory scenarios could most benefit from haptic augmentation. We're interested in applications not well-served by hearing aids / auditory amplification.

Psychophysical Evaluation